



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (17/08/2026 – 23/08/2026)

WEEK 4

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

**DESIGN AND IMPLEMENT GRAPHICAL USER
INTERFACE FOR AN OPEN SOURCE FIREWALL**

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COLLEGE NAME :

BITS PILANI

17/08/2026	- Started working on the prototype for the knowledge based chatbot. - Finalized the initial approach for handling unstructured document data.
18/08/2026	Worked on document loading, text preprocessing, and preparing the data for semantic search
19/08/2026	- Implemented text chunking and generated embeddings using a sentence transformer model. - Stored the embeddings in ChromaDB for semantic search.
20/08/2026	- Tested semantic retrieval with different queries. - Improved chunking and meta data handling for better retrieval results.
21/08/2026	- Learning more about chromaDB and retrieval of unstructured data. - Reviewed and refined the retrieval of unstructured data prototype.
22/08/2026	Holiday (2nd Saturday)
23/08/2026	Holiday (Sunday)

Next Week Plan	Evaluate the prototype using different types of queries.
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REFERENCE:

<https://docs.trychroma.com/docs/overview/introduction>

<https://sbert.net/>

<https://huggingface.co/sentence-transformers>

<https://www.youtube.com/watch?v=QojA5urIDzA>

<https://www.youtube.com/watch?v=JfSmffOyV-8>

SCREEN SHOTS:

```
import sys
import os

# Add src to path
sys.path.insert(0, os.path.join(os.path.dirname(__file__), 'src'))

from src.ingestion import run_ingestion
from src.cli import main as cli_main

def main():
    """Main entry point with command routing."""
    import argparse

    parser = argparse.ArgumentParser(description="Week 1: Unstructured Knowledge")
    parser.add_argument("command", choices=["ingest", "query", "eval"],
                        help="Command to run")
    parser.add_argument("--docs", default="data/documents", help="Documents directory")
    parser.add_argument("--db", default="chroma_db", help="ChromaDB persist directory")

    args = parser.parse_args()

    if args.command == "ingest":
        run_ingestion(documents_dir=args.docs, persist_dir=args.db)

    elif args.command == "query":
        # Update retriever to use correct paths
        os.chdir(os.path.dirname(__file__))
        cli_main()

    elif args.command == "eval":
        os.chdir(os.path.dirname(__file__))
        from evaluation.evaluate import run_evaluation
```

